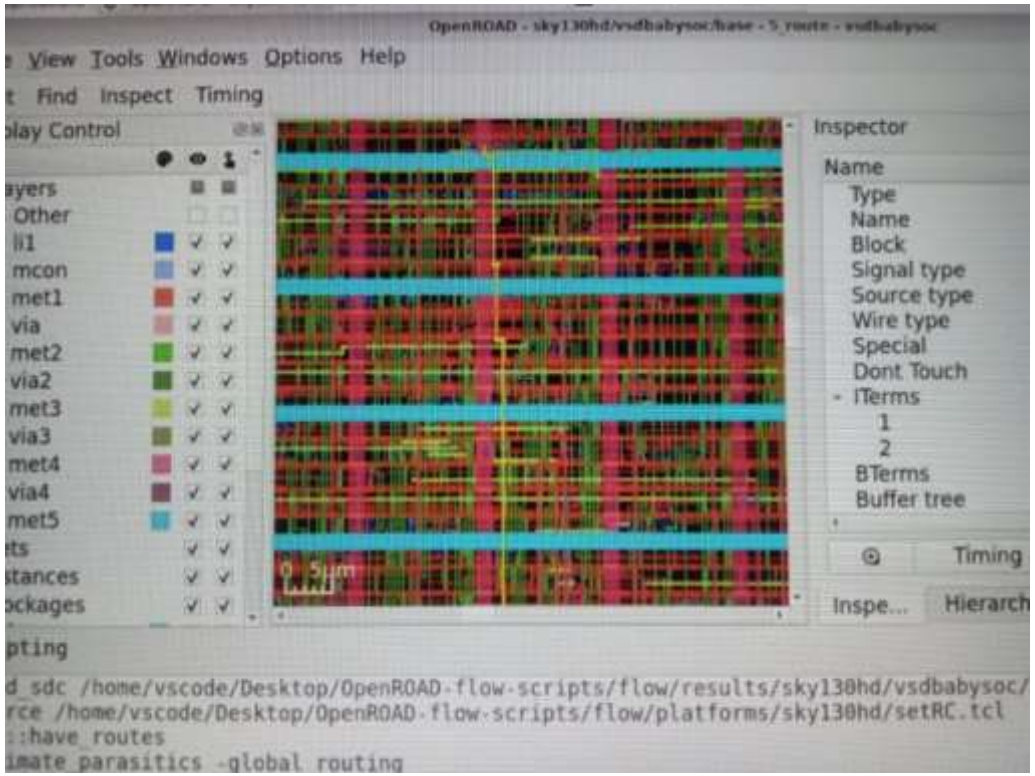
STA Reports



Static Timing Analysis (STA) is a method used to verify the timing performance of a digital circuit without applying input test vectors or simulating its functional operation. It checks whether signals propagate through the circuit within the required timing limits.

In this project, STA is performed on the VSDBabySoC design using the OpenROAD physical-design flow and SkyWater SKY130 High-Density standard-cell timing libraries. Multiple Process, Voltage, and Temperature (PVT) corners are considered to evaluate the timing behaviour under different operating conditions.

PVT stands for Process, Voltage, and Temperature. Variations in these parameters affect the delay of standard cells and the overall timing performance of a digital design.
FF (Fast-Fast): Represents fast-transistor process conditions, generally associated with shorter cell delays.
SS (Slow-Slow): Represents slow-transistor process conditions, generally associated with longer cell delays.
TT (Typical-Typical): Represents nominal process conditions.
Voltage and temperature variations: Different voltage and temperature combinations provide additional operating scenarios for timing verification.
Analysing multiple corners helps determine whether the design meets timing requirements across a range of operating conditions.

Setup Timing

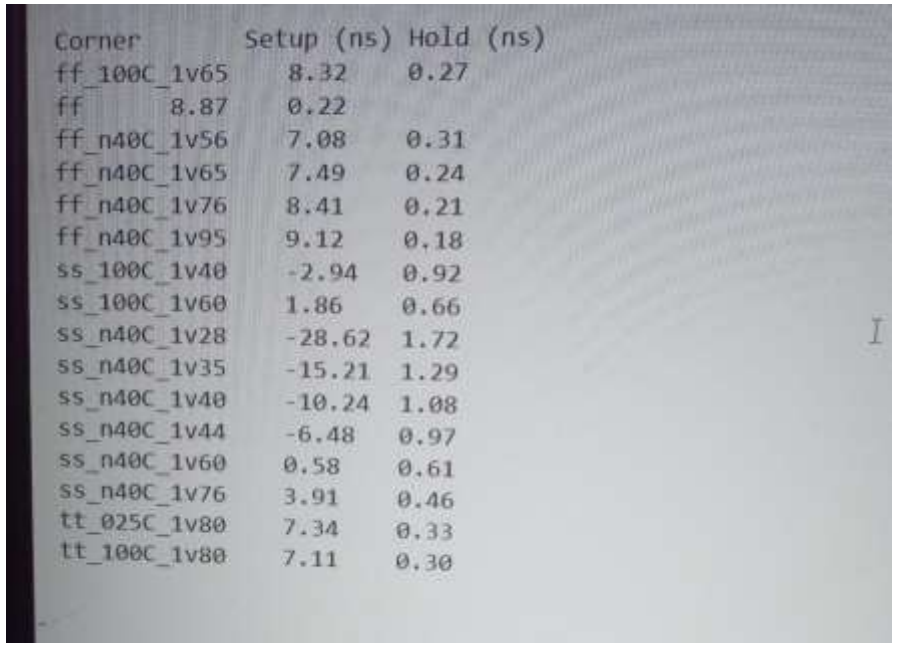
Setup timing verifies that data reaches a flip-flop input sufficiently before the active clock edge. A setup violation occurs when data arrives too late.

Positive setup slack indicates that the timing requirement is met, whereas negative setup slack indicates a setup timing violation.

Hold Timing

Hold timing verifies that data remains stable for the required duration after the active clock edge. A hold violation occurs when new data arrives too early.

Positive hold slack indicates that the hold requirement is met, whereas negative hold slack indicates a hold timing violation.



Corner	Setup (ns)	Hold (ns)
ff_100C_1v65	8.32	0.27
ff	8.87	0.22
ff_n40C_1v56	7.08	0.31
ff_n40C_1v65	7.49	0.24
ff_n40C_1v76	8.41	0.21
ff_n40C_1v95	9.12	0.18
ss_100C_1v40	-2.94	0.92
ss_100C_1v60	1.86	0.66
ss_n40C_1v28	-28.62	1.72
ss_n40C_1v35	-15.21	1.29
ss_n40C_1v40	-10.24	1.08
ss_n40C_1v44	-6.48	0.97
ss_n40C_1v60	0.58	0.61
ss_n40C_1v76	3.91	0.46
tt_025C_1v80	7.34	0.33
tt_100C_1v80	7.11	0.30